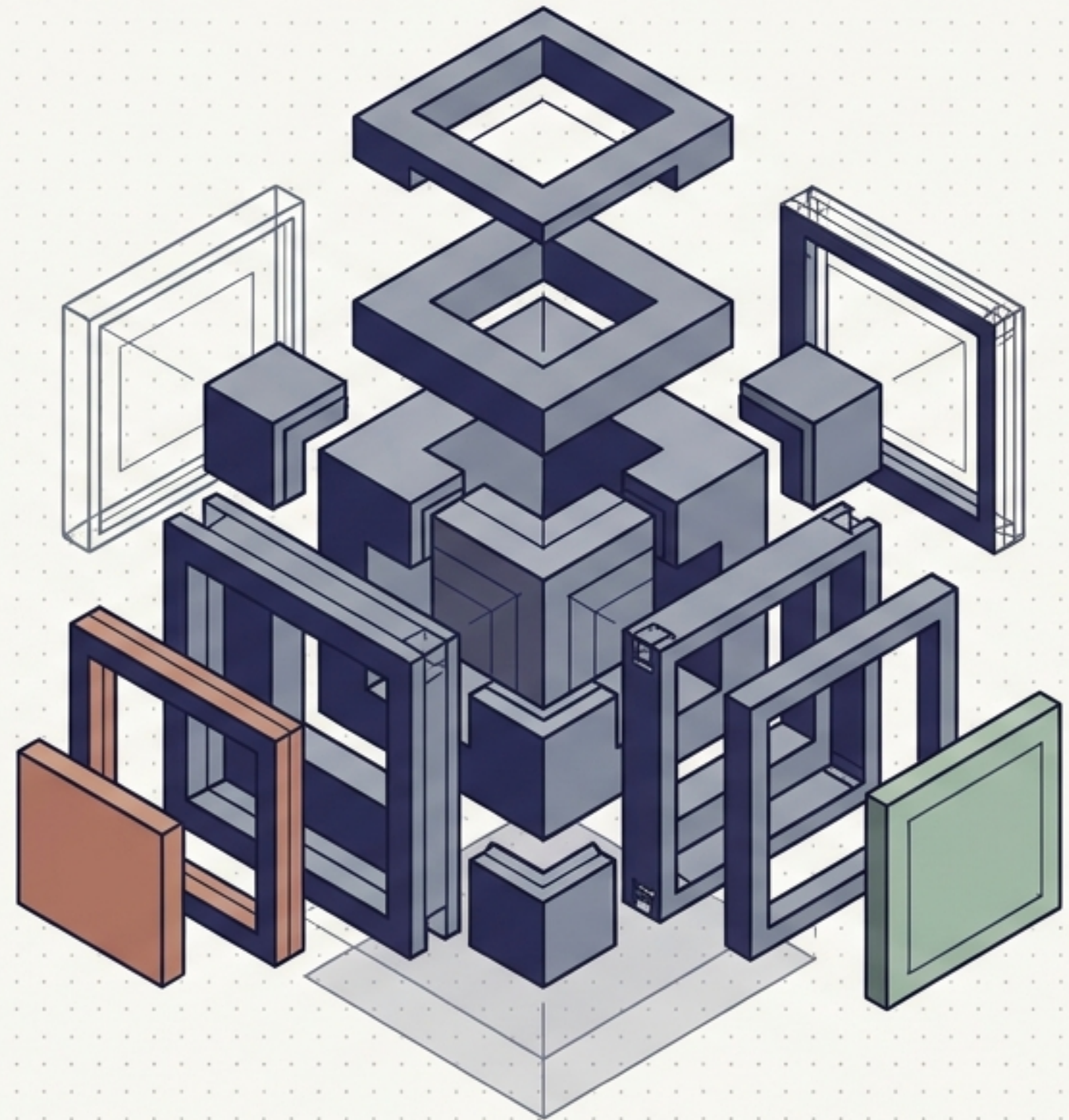


# From User to Architect: Building Custom GPTs

An executive syllabus for architecting intelligent, domain-specific AI assistants.

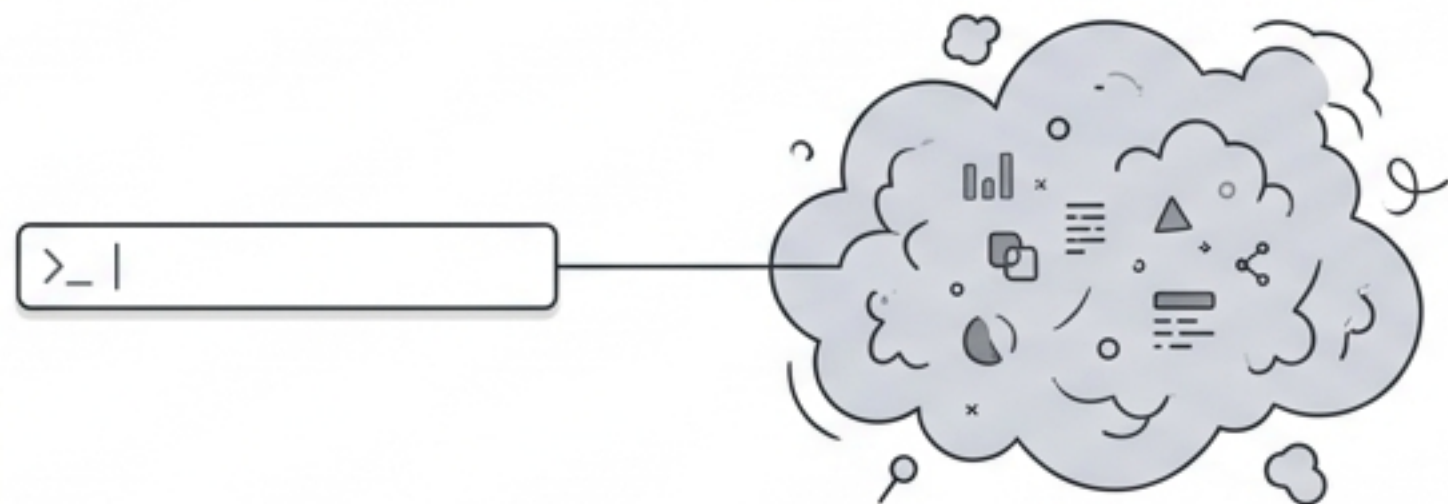
Featuring the curriculum by  
Vanderbilt University & Coursera





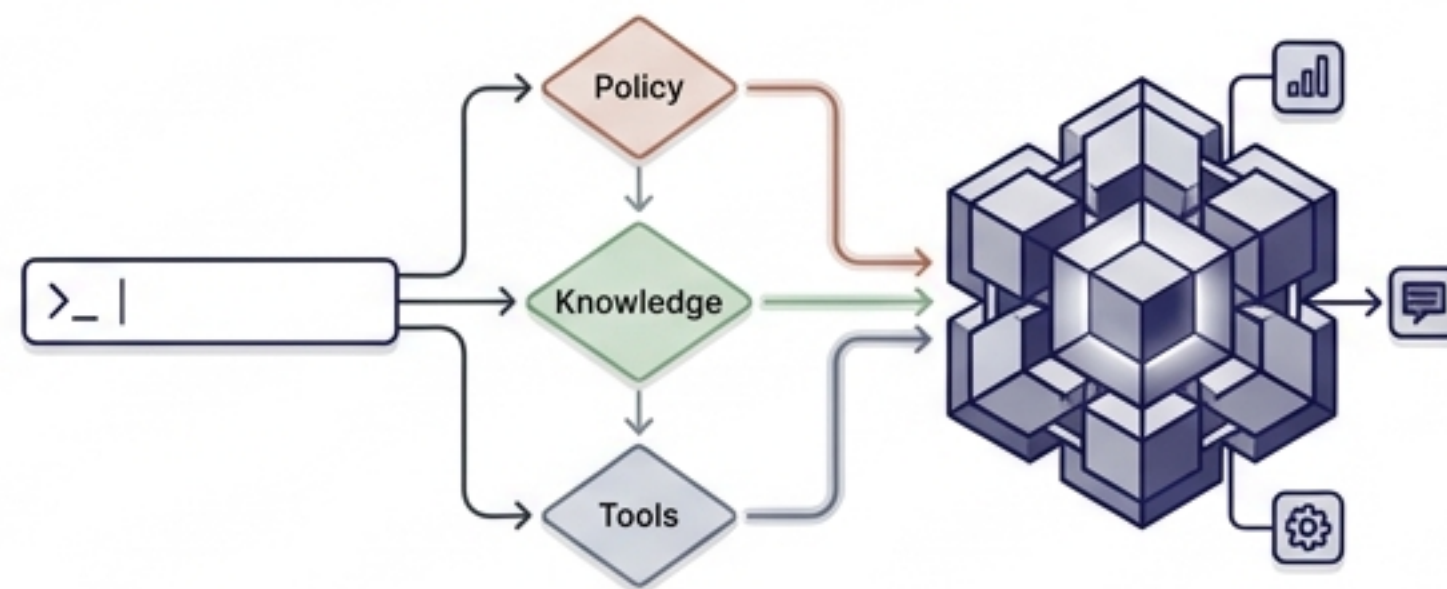
# The Evolution of Generative AI

## The Past: **Prompting**



One-size-fits-all, generic responses, unpredictable outputs.

## The Future: **Architecting**

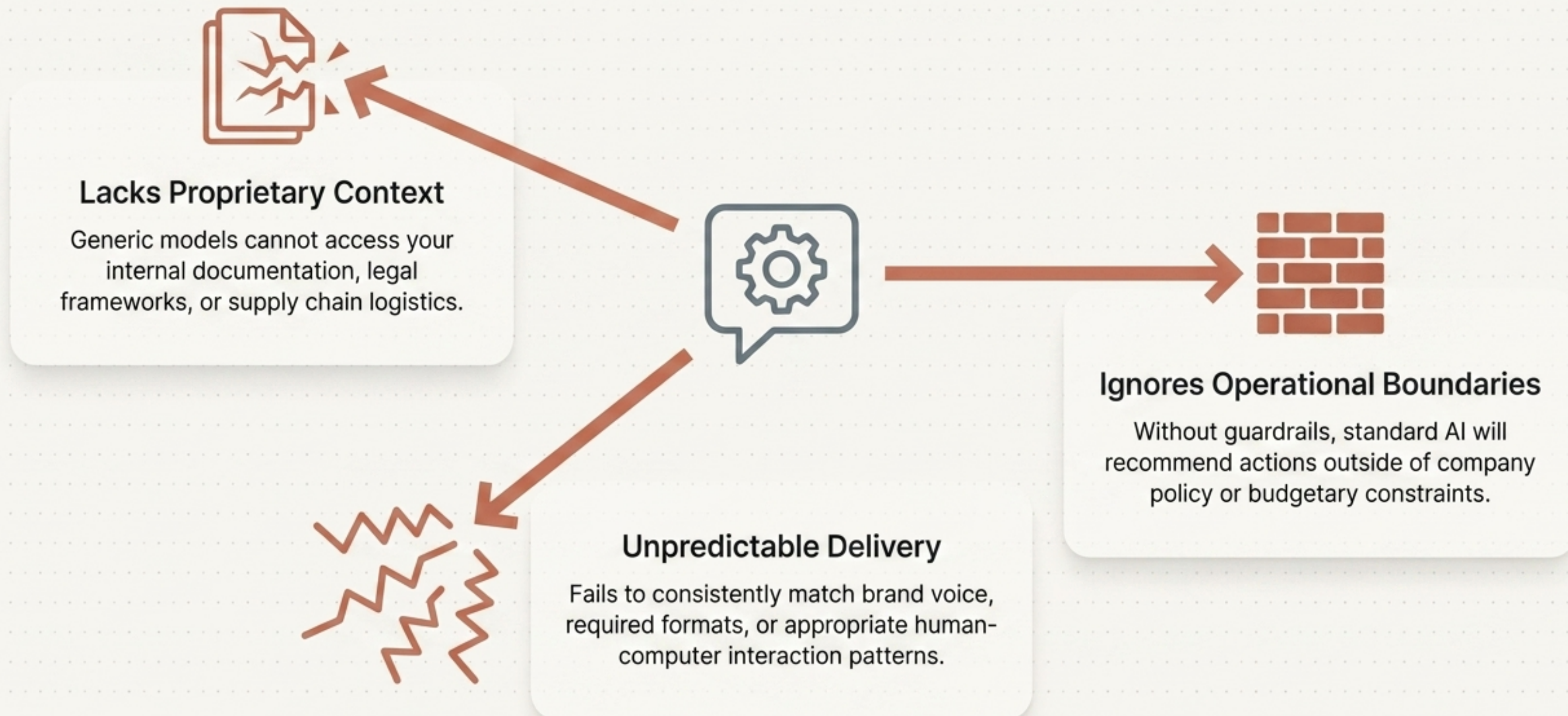


Specialized domain knowledge, rigid policy adherence, integrated automated actions.

**The competitive advantage has shifted** from typing prompts into a generic chatbox to **wiring intelligent systems** that **understand your exact domain**.

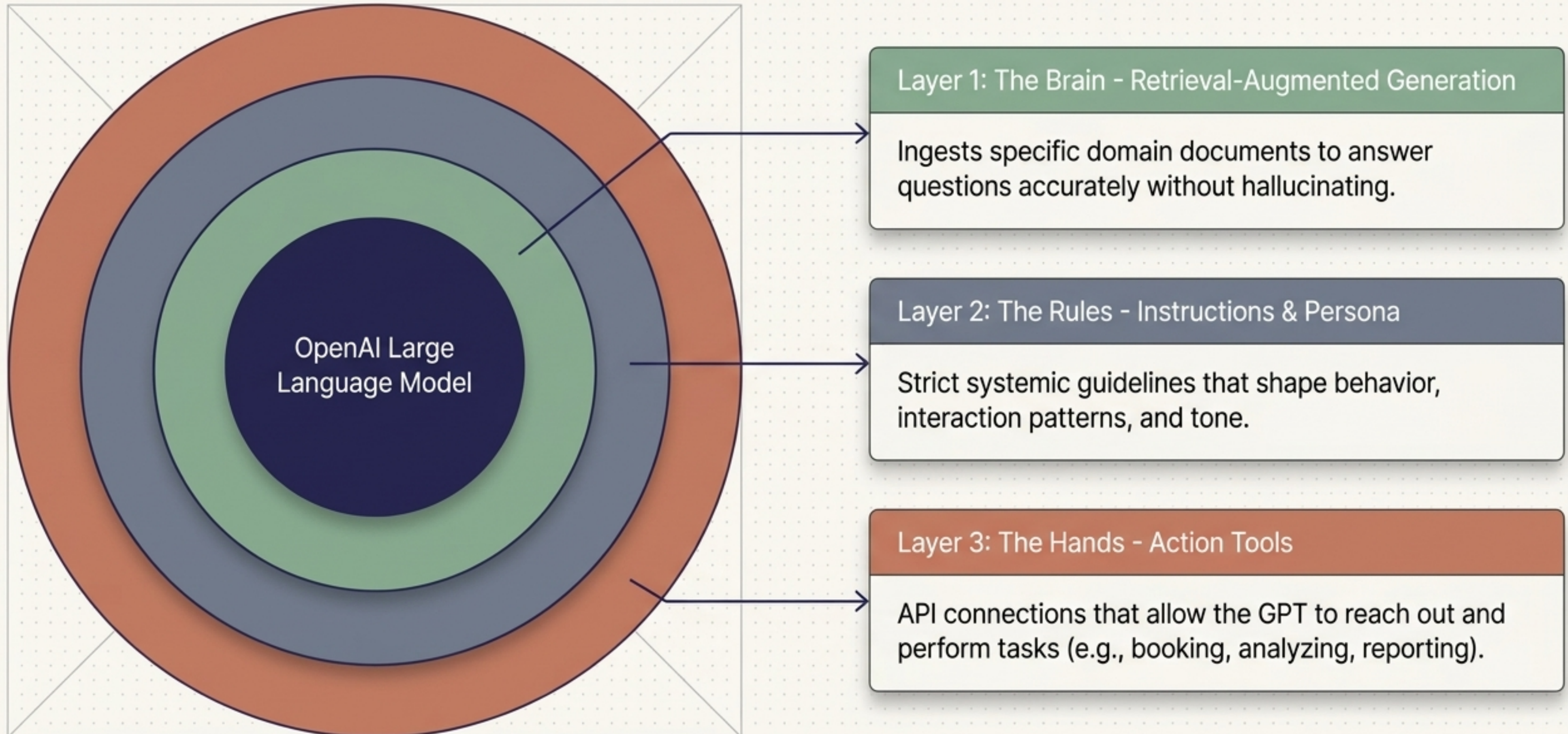


# The Generic AI Bottleneck



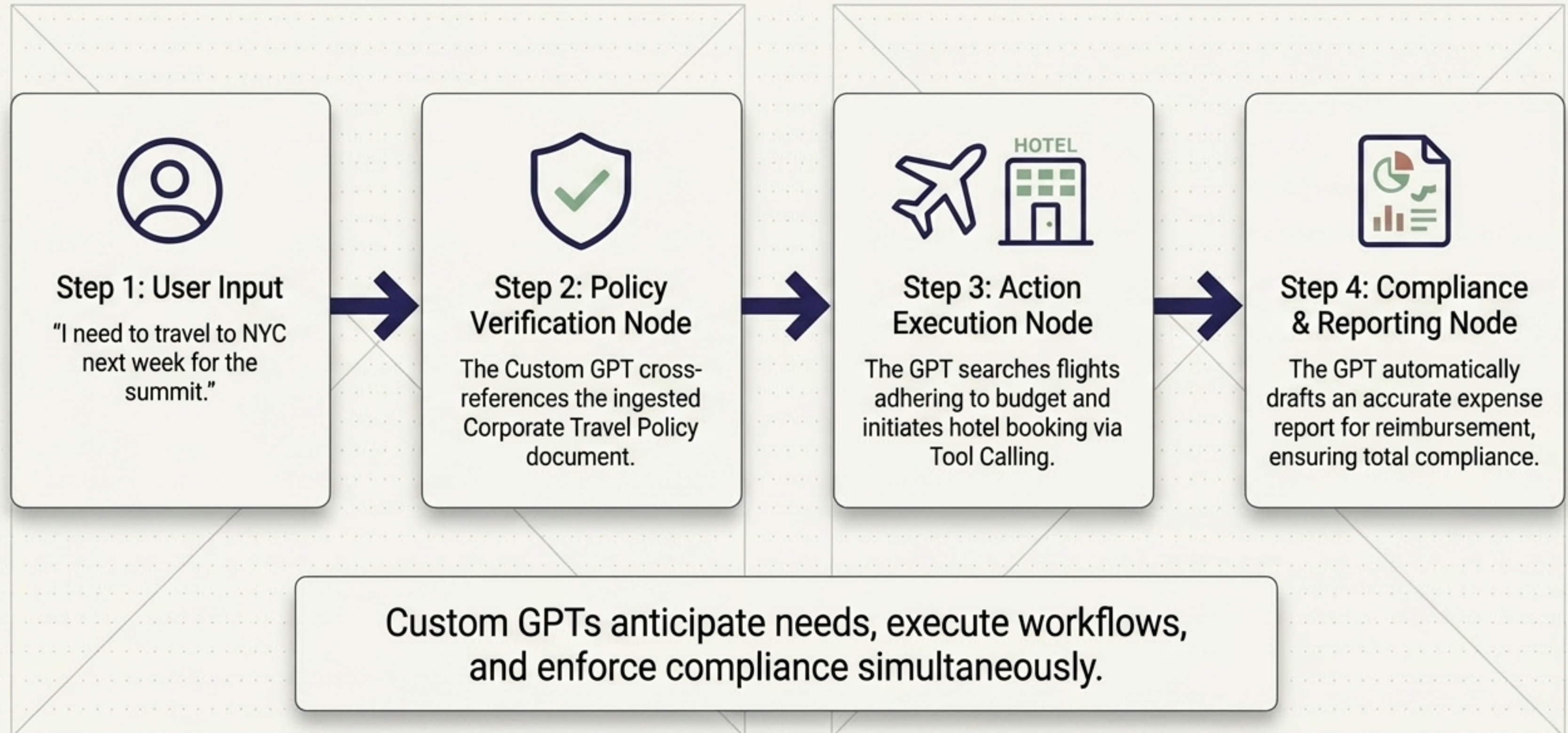


# The Anatomy of a Custom AI Assistant





# Real-World Architecture: Travel & Expense Management





# The Definitive Guide to AI Architecture



**Dr. Jules White**  
Vanderbilt University

Top Instructor

**79,713+**

Professionals currently enrolled.

**4.8/5 Stars**

Across 789 verified reviews.

**1.2 Million**

Total learners taught by Dr. White.

**Beginner  
Friendly**

No coding background required.

OpenAI GPTs: Creating Your Own Custom AI Assistants — available exclusively on Coursera.



# Beyond the Basic Setup

| Standard AI Courses              | The Vanderbilt Approach                                |
|----------------------------------|--------------------------------------------------------|
| Quick demos and prompt tweaking. | System evaluation and human-centered design. ✓         |
| Happy-path testing only.         | Adversarial inputs, boundaries, and ambiguous cases. ✓ |
| A novelty chatbot.               | A dependable, enterprise-ready AI assistant. ✓         |

**Building a GPT is easy. Designing a reliable, safe, and accurate GPT requires architectural discipline.**



# A Strategic Capability for Every Discipline



## **Business Leaders & Entrepreneurs**

Build agents to streamline supply chain logistics, ensure operational compliance, and unlock new realms of productivity.



## **Developers & Builders**

Master prompt patterns, tool calling, and retrieval-augmented generation to rapidly prototype intelligent systems.



## **Educators & Analysts**

Deploy custom GPTs for complex legal document analysis, scientific hypothesis generation, and personalized learning.

Zero coding required. The curriculum focuses on logic, design, and system evaluation.



# The 7-Hour Architectural Transformation

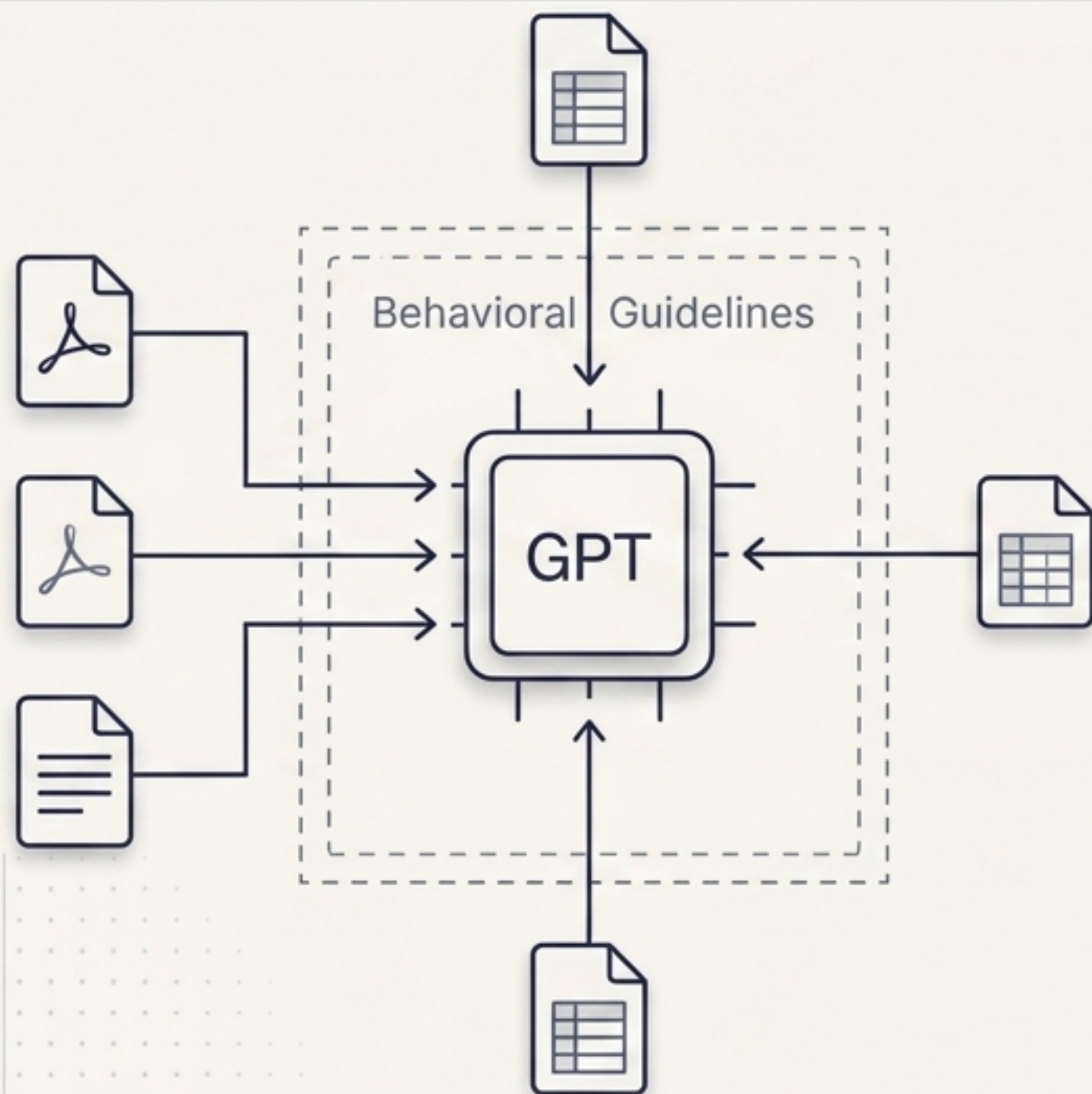


A flexible, self-paced curriculum featuring guided hands-on practice, AI-graded assignments, and structured reading.



# Module 1: Foundational Architecture

2 Hours



## Knowledge Ingestion

Bringing in outside information to eliminate hallucinations.

ON ☒

## Behavioral Shaping

Defining the specific task, scope, and limitations of the assistant.

ON ☒

## Setup Choices

Navigating the foundational architecture of the OpenAI ecosystem.

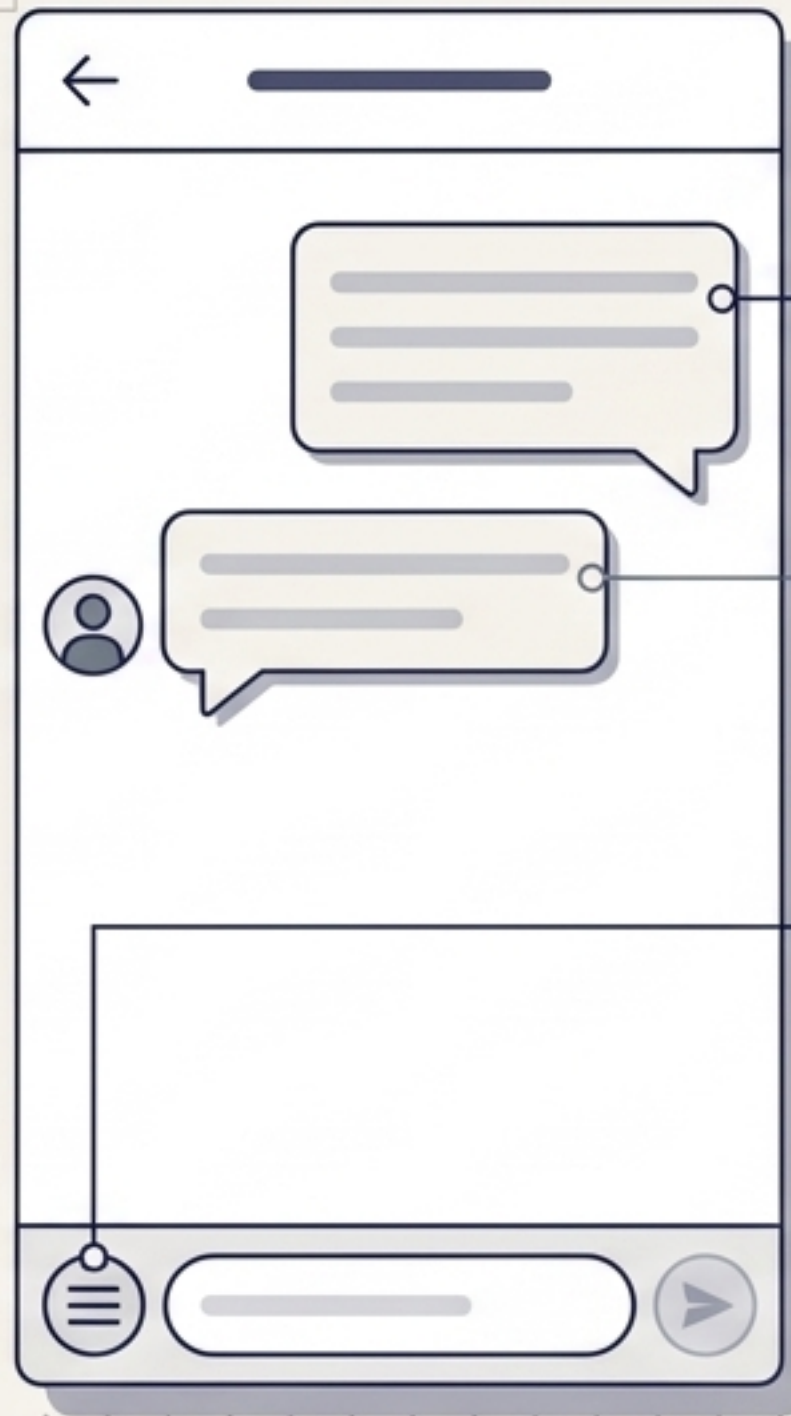
ON ☒

**Takeaway:** Establish the core brain and operational boundaries of your custom assistant.



# Module 2: Interaction Design & Patterns

3 Hours



## Prompt Patterns

Advanced frameworks for structuring outputs and guiding user queries.

## Persona Design

Customizing the tone and communication style of your GPT.

## Human-Computer Interaction (HCI)

Designing menu actions and conversational flows for seamless user experiences.

## Takeaway:

Move beyond basic chatting to engineering deliberate, highly effective human-AI interactions.

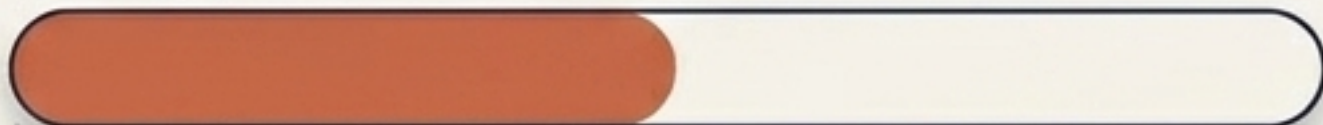


# Module 3: Rigorous System Evaluation 2 Hours

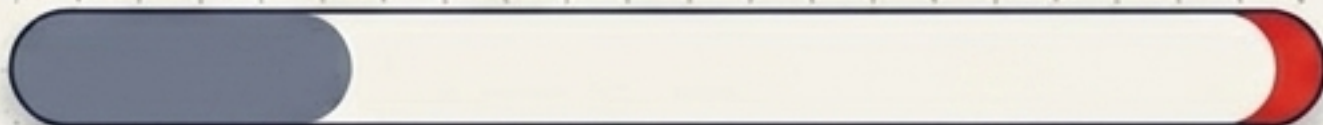
Expected Cases



Ambiguous Cases



High-Risk Situations



## Test Case Generation

Building benchmarks to evaluate how a GPT performs under pressure.

## Edge-Case Handling

Designing responses for ambiguous, high-risk, or adversarial inputs.

## Safer Responses

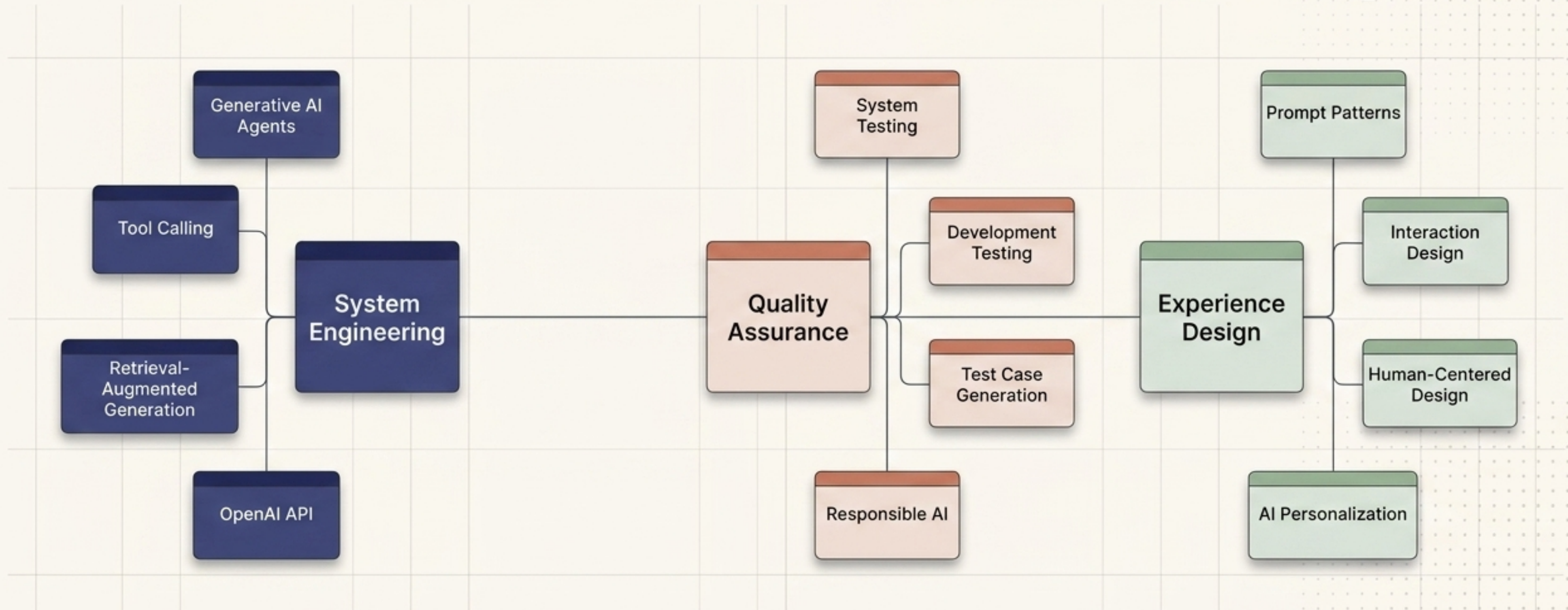
Enforcing strict adherence to corporate safety and brand guidelines.

## Takeaway:

Ensure your AI assistant is accurate, reliable, and safe for real-world deployment.



# The Architect's Skill Taxonomy





# Verifiable Expertise & ROI

## Learner Testimonials

“Outstanding course material... These skills are the next core expectation for all corporate jobs in the future.” – DS

“I directly applied the concepts and skills to an exciting new project at work.” – Jennifer J.

## The Credential



- **Earn a Career Certificate:** Add this verified Vanderbilt University credential directly to your LinkedIn profile, resume, or CV.
- **Immediate Application:** 97% of learners liked the course; apply skills to your job on day one.



# Deploy Your First Custom Agent

The era of generic AI is over. Start building intelligent systems tailored to your unique challenges today.

## For Individuals

- Enroll today on Coursera.
- Complete in 7 hours.
- Earn your Vanderbilt Certificate.

[Start Learning →](#)

## For Businesses & Teams

- Upskill your entire workforce.
- Unlock Coursera for Business.
- Scale custom AI architecture across your enterprise.

[Learn More →](#)

Search "OpenAI GPTs: Creating Your Own Custom AI Assistants" on Coursera.org to begin.